



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: EEE G511
Course Title	: RF AND MICROWAVE CIRCUIT DESIGN
Instructor-in-Charge	: Dinesh Rano
Instructor(s)	: None
Practical Instructors	: Laveena Dhadich (RS)

1. Course Description:

Introduction to radio frequency engineering; advantages; various frequency bands; applications; Microwave passive components and high frequency modeling; Microwave network analysis: scattering parameters and Signal flow graph; vector network analyzer-dynamic range and IF bandwidth; Transmission lines; microstrip lines; impedance matching techniques-transmission line based and lumped circuit approach and layout generation. Power dividers and couplers, RF filter design; Filter scaling and transformation; Filter implementation using microstrip lines and lumped components; Amplifier design; noise, harmonic distortion, and intercept point analysis, amplifier design for gain and noise figure; oscillator design, one-port and two-port oscillator design, phase noise of oscillator, introduction to EMI and EMC, recent trends.

2. Scope and Objective of the Course:

Identify basic RF devices/components and their applications, understand common rules of thumb in RF design, use scattering matrix for RF network analysis, design planar transmission lines, design impedance-matching circuits, use the Smith chart for impedance matching, implement passive RF devices in planar technology, design passive RF devices with RF CAD tools, learn the use of a vector network analyzer for RF measurements, learn the basic principles of RF amplifier

3. Course learning outcomes:

- Network analysis – two, three, and four-port networks with SFG and VNA
- Transmission lines – microstrip line design and synthesis
- Impedance matching network – Design of single, dual, and broadband matching for RF front-end
- Microwave components design
- Amplifiers and RF filters design

4. Text Book:

| David M. Pozar, Microwave Engineering, 4th edition. Wiley, 2015

Reference Books:

| R.E. Collin, Foundations for Microwave Engineering, 2nd edition. Wiley, 2007
| A. Das and S. K. Das, Microwave Engineering, 3rd edition., McGraw Hill Edu., 2015

5. Method of conducting the course:

Traditional classroom lectures – involvement of the board and slides for understanding. A 2-Hour Weekly lab session will be conducted. This will be taught by the lab instructor. Attendance is mandatory for the lab session.



6. Course Plan:

Module No.	Lecture No. / Topic	Reference	Learning outcomes
1	1-3 Introduction to radio frequency engineering	Slides	Introduction to radio frequency engineering - advantages, various frequency bands - applications, Microwave passive components, and high frequency modeling of RF circuits - resistors, capacitors, and inductors, Q-factor, bandwidth.
2	4-10 Network analysis	TB – Ch-4 (4.2, 4.3, 4.5), Slides	Microwave network analysis - Multi-port Networks, Z-, Y-, and S-Matrix, scattering parameters - two, three, and four port analysis, matched/lossless/reciprocal Networks, signal flow graph, introduction to vector network analyzer - calibration and measurement, dynamic range, and IF stages and bandwidth.
3	11-17 Transmission Lines	TB – Ch-3 (3.1, 3.8, 3.10, 3.11), slides	Microstrip lines, quasi-TEM mode, design aspects, phase and group velocity, dispersion (briefly), frequency-dependent effects and higher-order modes, design of frequency-dependent loads (FDCLs) with lumped elements
4	18-25 Impedance Matching	TB – Ch-5 (5.1 – 5.8), slides	Impedance matching techniques - transmission line-based (Smith chart, multi-stage quarter wave transformer, reflection analysis), Monzon network (R-R matching), and C-R and R-C matching and lumped circuit approach - L-, T-, Pi- matching circuit, and layout generation, matched/lossless/reciprocal networks, broadband matching technique
5	26 – 32 Microwave components (power divider, coupler, isolator, circulator, RF filter)	TB – Ch-7 and 8 (7.1 – 7.8, 8.2 – 8.5), slides	Power dividers and couplers, basic Properties, PD-design aspects, even and odd mode analysis, Wilkinson Power divider, quadrature coupler, coupled-line coupler, RF filter design, Filter Phase Function, The Linear Phase Filter, Insertion Loss Method, Filter scaling and transformation, Stepped Impedance Method, Filter implementation using microstrip lines and lumped components, Richard's transformation and Kuroda identities
6	33-36 RF amplifiers and oscillators	TB – Ch-10, 12, and 13 (10.1 – 10.4, 12.1, 12.2, 12.5 and 13.1 – 13.3), slides	Noise analysis – Noise figure (NF), NF of cascaded system, harmonic and intermodulation distortion of RF circuits, and intercept point analysis – 3 rd order, passive intermodulation (PIM), amplifier design – two-port power gain, stability circle, power amplifier, gain and noise figure of LNA, Oscillator design, one-port and two-port oscillator design, phase noise of oscillator
7	36-40 Introduction to EMI and EMC	Slides	EMC standards, signal spectrum, crosstalk, coupling effects, shielding for near and far-field sources



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7. Evaluation Scheme:

Component	Duration	Weightage (%)	Marks	Date of evaluation	Nature of component
Quiz (4 quizzes during lecture hours – all compulsory)	15 Min.	10	10	TBA	closed book
Mid-semester test	90 Min.	20	20	07/10, FN2	Closed book
Laboratory performance and assignment – 2 (seminar presentation in group/individual – 2 + simulation project)	TBA	40	40	TBA in class	Open book
Comprehensive Examination	3 hrs	30	30	08/12, AN	Closed and open book
Total		100			

Mid-semester grading (total = 40 marks): Mid-sem test (20 marks) + Quiz (5 marks) + Lab + Assignment (15 marks)

8. Chamber Consultation Hour: To be announced by I/C

9. Notices: Notices will be displayed only on *Nalanda* portal.

10. Make-up Policy: Makeup will not be given for quizzes or assignments. Make-up for other components is a one-time provision that applies only in cases of "extreme medical emergency". Seasonal fever, allergy, loose motion, vomiting, mood swing, attending celebrations, ceremonies etc do not qualify for make-up.

11. RF Lab Session: Lab will be performed on an RF design tool such as CST/ADS/Sonnet. Students are required to install the free version on their laptops and then register online.

Instructions

- | The Quiz (short test) will be conducted during the lecture hour, which will be announced a week earlier
- | Students will perform the Lab simulations individually.
- | Simulation problems for practice will be uploaded in *Nalanda*. Students will maintain a lab record of the simulations, with the write-up of Objectives, Principles, Design, CAD model, Results and Conclusion. The record will be evaluated thoroughly for its originality, correctness, technical integrity, and neatness.
- | Students' RF design skills will be tested by assigning design problems, which will be rigorously evaluated.

Instructor-in-charge
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END



FIRST SEMESTER 2026-27

Course No : **EEE_G511**
Course Title : **RF AND MICROWAVE CIRCUIT DESIGN LAB**
Instructor-in-Charge : **Dinesh Rano**
Lab Instructor : **Laveena Dadhich (RS)**

List of Experiments

- 1. Design and analysis of microstrip lines:**
 - a. Varying width (W) and substrates (ϵ_r) $\rightarrow$ plot Z_o vs W for different ϵ_r (at least 10 points)**
 - b. Design load terminations: matched/unmatched with those of the microstrip line's characteristic impedance $\rightarrow$ port and discrete impedance**
- 2. Emulation of inductive and capacitive loads with TX line (CST) and matching with stub for:**
 - a. Real-to-real termination**
 - b. Real to complex termination**
- 3. a. VNA calibration: Through, Open, Short, and Load termination.**
b. Determination of the S-matrix of a transmission line (2-port) with VNA and validation with theory
- 4. Design and development of matching networks (CST):**
 - a. Quarter wave (single section and multi-section) – single band**
 - b. Dual band**
 - c. Tri-band**
- 5. Design of planar microwave devices: directional coupler, attenuator, Wilkinson power divider, circulators**
- 6. Measurement of power received for an antenna setup with VNA and a Spectrum analyzer:**
 - a. Horn antenna**
 - b. Patch antenna**
- 7. Familiarization with Microwave Bench Setup and equipment: Characterization of Reflex Klystron Modes**
- 8. Determination of VSWR, Reflection Coefficient, and Frequency of Operation with Reflex Klystron**

Dinesh Rano